NEET TEST HUB

NTH-PREMIUM TEST SERIES 11

Date – 07 December 2025	Duration – 3 Hours	Marks – 720

Physics :- Electric Charges & Fields, Electrostatic Potential & Capacitance, Current Electricity

Chemistry :- Solutions, Electrochemistry, Chemical Kinetics

Biology :- Sexual Reproduction in Flowering Plants, Human Reproduction, Reproductive Health

Instruction

1. The test is of 3 hours duration.

ions. The maximum mark is 720.

n Paper, Sections I, II, III, and IV consisting
istry), Section III (Botany) and Section IV
bject and each subject is divided into two
Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. Two wires A and B of the same material, having radii in the ratio $1:2$ carry currents in the ratio $4:1$. The ratio of drift speed of electrons in A and B is

A) $16:1$ **B)** $1:16$ **C)** $1:4$ **D)** $4:1$

2. The total momentum of electrons in a straight wire of copper of length $1\ metre$ carrying a current of $16\,A$ is

A) $91 \times 10^{-12}\,kg\,m/sec$ **B)** $91 \times 10^{-15}\,kg\,m/sec$
C) $91 \times 10^{-14}\,kg\,m/sec$ **D)** $91 \times 10^{-6}\,kg\,m/sec$

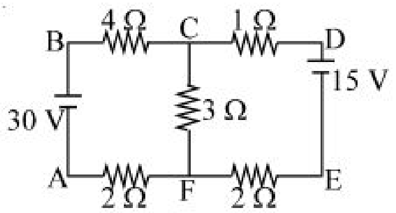
3. A meta sample carrying a current along $x-$ axis with density J is subjected to a magnetic field B (along $z-$ axis). The electric field E developed along $y-$ axis is directly proportional to J as well as B. The constant of proportionality has $SI\ unit$

A) C/m^2 **B)** m^2s/C
C) m^2/C **D)** m^3/C

4. The figure shows a network of resistors and a battery. If $1\ A$ current flowsthrough the branch CF, then answer the following questions The

A) branch DE is $1\,A$
B) branch BC is $2\,A$
C) branch BG is $4\,A$
D) Both (A) and (B)

5. The figure shows a network of five resistances and two batteries The current through the $15\,V$ battery is .... A



A) 0 **B)** 1 **C)** 3 **D)** none

6. An electric kettle has two heating coils. When one of the coils is connected to an a.c. sotuce, the water in the kettle boils in 10 minutes. When the other coil is used the w'ater boils in 40 minutes. If both the coils are connected in parallel, the time taken by the same quantity of water to boil will be ...... min

A) 15 **B)** 25 **C)** 8 **D)** 4

7. A battery of $e.m.f.\ 10\,V$ and internal resistance $0.5\,ohm$ is connected across a variable resistance R. The value of R for which the power delivered in it is maximum is given by .... ohm

A) 2 **B)** 0.25 **C)** 1 **D)** 0.5

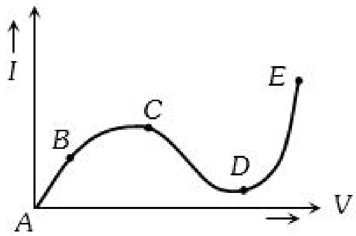
8. Thomson coefficient of a conductor is $10\,\mu V/K$. The two ends of it are kept at $50\,^oC$ and $60\,^oC$ respectively. Amount of heat absorbed by the conductor when a charge of $10\,C$ flows through it is

A) $1000\,J$ **B)** $100\,J$
C) $100\,mJ$ **D)** $1\,mJ$

9. What is immaterial for an electric fuse wire

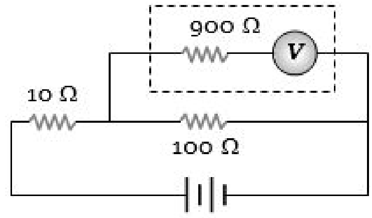
A) Its specific resistance **B)** Its radius
C) Its length **D)** Current flowing through it

10. From the graph between current I and voltage V shown below, identify the portion corresponding to negative resistance



A) AB **B)** BC **C)** CD **D)** DE

11. The potential difference across the $100\ \Omega$ resistance in the following circuit is measured by a voltmeter of $900\ \Omega$ resistance. The percentage error made in reading the potential difference is

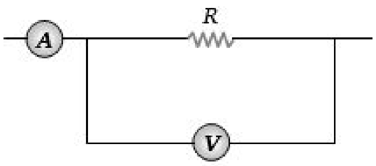


A) $\frac{10}{9}$ **B)** 0.1 **C)** 1 **D)** 10

12. A Daniel cell is balanced on $125\ cm$ length of a potentiometer wire. Now the cell is short-circuited by a resistance $2\,ohm$ and the balance is obtained at $100\,cm$. The internal resistance of the Daniel cell is .... ohm

A) 0.5 **B)** 1.5 **C)** 1.25 **D)** 0.8

13. The ammeter A reads $2\ A$ and the voltmeter V reads $20\ V$. the value of resistance R is (Assuming finite resistance's of ammeter and voltmeter)



A) Exactly $10\,ohm$
B) Less than $10\,ohm$
C) More than $10\,ohm$
D) We cannot definitely say

14. In an electrolyte 3.2×10^{18} bivalent positive ions drift to the right per second while 3.6×10^{18} monovalent negative ions drift to the left per second. Then the current is

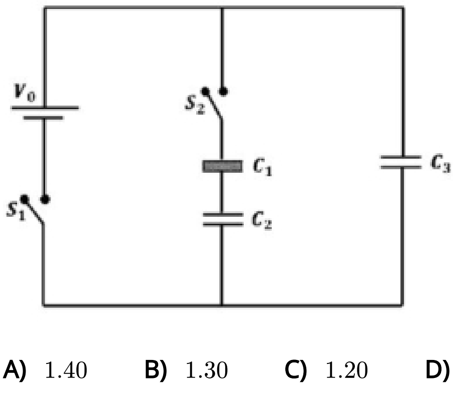
A) $1.6\,amp$ to the left
B) $1.6\,amp$ to the right
C) $0.45\,amp$ to the right
D) $0.45\,amp$ to the left

15. Current of $4.8\ A$ is flowing through a conductor. The number of electrons per second will be

A) 3×10^{19} **B)** 7.68×10^{21}
C) 7.68×10^{20} **D)** 3×10^{20}

16. . Three identical capacitors C_1, C_2 and C_3 have a capacitance of $1.0\mu F$ each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled

switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be $5\mu C$. The value of $\varepsilon_r = \ldots..$



A) 1.40 **B)** 1.30 **C)** 1.20 **D)** 1.50

17. Considering a group of positive charges, which of the following statements is correct?

A) Net potential of the system cannot be zero at a point but net electric field can be zero at that point.
B) Net potential of the system at a point can be zero but net electric field can't be zero at that point.
C) Both the net potential and the net field can be zero at a point.
D) Both the net potential and the net electric field cannot be zero at a point.

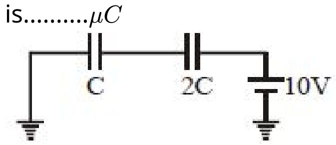
18. Which of the following statements is true about the flow of electrons in an electric circuit?

A) Electrons always flow from lower to higher potential
B) Electrons always flow from higher to lower potential
C) Electrons flow from lower to higher potential, except through power sources
D) Electrons flow from higher to lower potential, except through power sources

19. Assertion: Electron move away from a region of higher potential to a region of lower potential.
Reason: An electron has a negative charge.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
C) If the Assertion is correct but Reason is incorrect.
D) If the Assertion is incorrect but the Reason is correct.

20. In the circuit shown in the figure, $C = 6\ \mu F$. The charge stored in the capacitor of capacity C is..........μC

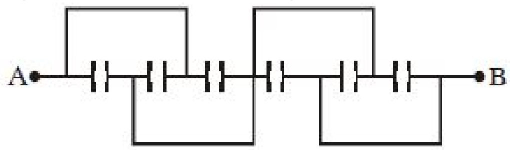


A) 0 **B)** 90 **C)** 40 **D)** 60

21. If an electron enters into a space between the plates of a parallel plate capacitor at an angle α with the plates and leaves at an angle β to the plates, the ratio of its kinetic energy while entering the capacitor to that while leaving will be

A) $(\cos\alpha/\cos\beta)^2$
B) $(\cos\beta/\cos\alpha)^2$
C) $(\sin\alpha/\sin\beta)^2$
D) $(\sin\beta/\sin\alpha)^2$

22. All capacitors used in the diagram are identical and each is of capacitance C. Then the effective capacitance between the points A and B is

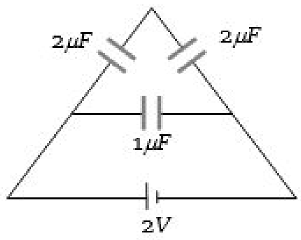


A) $1.5\,C$ **B)** $6\,C$ **C)** C **D)** $3\,C$

23. A charge Q is distributed over three concentric spherical shell of radii $a, b, c (a < b < c)$ such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be

A) $\frac{Q}{12\pi \in_0} \frac{ab+bc+ca}{abc}$
B) $\frac{Q\,(a^2+b^2+c^2)}{4\pi \in_0 (a^3+b^3+c^3)}$
C) $\frac{Q}{4\pi \in_0 (a+b+c)}$
D) $\frac{Q\,(a+b+c)}{4\pi \in_0 (a^2+b^2+c^2)}$

24. The charge on any one of the $2\,\mu\,F$ capacitors and $1\,\mu\,F$ capacitor will be given respectively (in $\mu\,C$) as



A) $1, 2$ **B)** $2, 1$ **C)** $1, 1$ **D)** $2, 2$

25. A hollow metallic sphere of radius R is given a charge Q. Then the potential at the centre is

A) Zero
B) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{R}$
C) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{2Q}{R}$
D) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{2R}$

A) $\sqrt{R_1} : \sqrt{R_2}$ **B)** $R_1 : R_2$
C) $R_1^2 : R_2^2$ **D)** $R_1^3 : R_2^3$

27. The potential gradient at which the dielectric of a condenser just gets punctured is called

A) Dielectric constant
B) Dielectric strength
C) Dielectric resistance
D) Dielectric number

28. Two electric charges $12\,\mu C$ and $-6\,\mu C$ are placed $20\,cm$ apart in air. There will be a point P on the line joining these charges and outside the region between them, at which the electric potential is zero. The distance of P from $-6\,\mu C$ charge is....... m

A) 0.10 **B)** 0.15 **C)** 0.20 **D)** 0.25

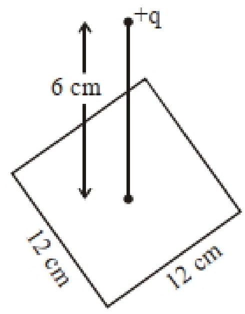
29. A conductor with a positive charge

A) Is always at $+ve$ potential
B) Is always at zero potential
C) Is always at negative potential
D) May be at $+ve$, zero or $-ve$ potential

30. A proton is accelerated through $50,000\ V$. Its energy will increase by

A) $5000\,eV$ **B)** $8 \times 10^{-15}\,J$
C) $5000\,J$ **D)** $50,000\,J$

31. A point charge of $+\,12\,\mu C$ is at a distance $6\ cm$ vertically above the centre of a square of side $12\ cm$ as shown in figure. The magnitude of the electric flux through the square will be .... $\times 10^3\ Nm^2/C$



A) 452 **B)** 381 **C)** 226 **D)** 113

32. The electric field in a region is given $\vec{E} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j}\right)\frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area $0.2\ m^2$ (parallel to $y-z$ plane) to that of the surface of area $0.3\ m^2$ (parallel to $x-z$ plane) is $a : b$, where $a =$ ....
[Here $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along x, y and $z-$ axes respectively]

A) 2 **B)** 3 **C)** 4 **D)** 1

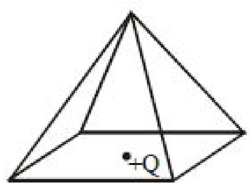
33. A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R. The force of gravity acts downwards. The equilibrium of the particle will be stable R

A) for all values of H
B) only if $H > \frac{R}{\sqrt{2}}$
C) only if $H < \frac{R}{\sqrt{2}}$
D) only if $H = \frac{R}{\sqrt{2}}$

34. If a body is charged by rubbing it. Its weight

A) always decreases slightly
B) always increases slightly
C) may increase slightly or may decrease slightly
D) remains precisely the same

35. A point charge $+Q$ is positioned at the centre of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is



A) $\frac{Q}{16\varepsilon_0}$ **B)** $\frac{Q}{4\varepsilon_0}$
C) $\frac{Q}{8\varepsilon_0}$ **D)** None of these

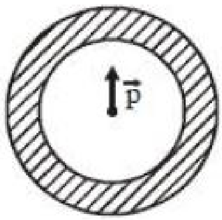
36. The point charges Q and $-2Q$ are placed at some distance apart. If the electric field at the location of Q is $\vec{E}$, then the electric field at the location of $-2Q$ will be :

A) $-\frac{\vec{E}}{2}$ **B)** $-\frac{3\vec{E}}{2}$
C) $+\frac{\vec{E}}{2}$ **D)** $-2\,\vec{E}$

37. Let $\rho(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'p' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is

A) 0
B) $\frac{Q}{4\pi\varepsilon_0 r_1{}^2}$
C) $\frac{Q}{4\pi\varepsilon_0 R^4}$
D) $\frac{Q r_1{}^2}{3\pi\varepsilon_0 R^4}$

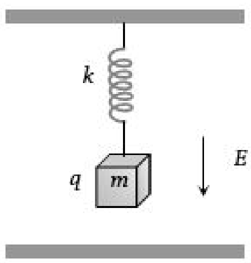
38. Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b, and carries charge Q. At its centre is a dipole $\vec{P}$ as shown. In this case



A) Surface charge density on the inner surface of the shell is zero everywhere
B) Electric field outside the shell is the same as

uniform and equal to $\frac{[illegible]}{4\pi a^2}$
D) Surface charge density on the outer surface depends on $\left|\vec{P}\right|$

39. Time period of a block suspended from the upper plate of a parallel plate capacitor by a spring of stiffness k is T. When block is uncharged. If a charge q is given to the block them, the new time period of oscillation will be



A) T **B)** $> T$
C) $< T$ **D)** $\geq T$

40. The unit of electric permittivity is

A) $Volt/m^2$ **B)** $Joule/coulomb$
C) $Farad/m$ **D)** $Henry/m$

41. One metallic sphere A is given positive charge whereas another identical metallic sphere B of exactly same mass as of A is given equal amount of negative charge. Then

A) Mass of A and mass of B still remain equal
B) Mass of A increases
C) Mass of B decreases
D) Mass of B increases

42. A charge q_1 exerts some force on a second charge q_2. If third charge q_3 is brought near, the force of q_1 exerted on q_2

A) Decreases
B) Increases
C) Remains unchanged
D) Increases if q_3 is of the same sign as q_1 and decreases if q_3 is of opposite sign

43. When a body is earth connected, electrons from the earth flow into the body. This means the body is.....

A) Unchanged **B)** Charged positively
C) Charged negatively **D)** An insulator

44. An electron and a proton are in a uniform electric field, the ratio of their accelerations will be

A) Zero
B) Unity
C) The ratio of the masses of proton and electron
D) The ratio of the masses of electron and proton

45. When a glass rod is rubbed with silk, it

A) Gains electrons from silk
B) Gives electrons to silk
C) Gains protons from silk
D) Gives protons to silk

Chemistry

46. What is the two third life of a first order reaction having $\mathrm{k} = 5.48 \times 10^{-14}\mathrm{sec}^{-1}$?

A) 2.01×10^{11} s
B) 2.01×10^{13} s
C) 8.08×10^{13} s
D) 16.04×10^{11} s

47.

The half-life for decay of radioactive $^{14}\mathrm{C}$ is 5730 years. An archaeological artefact containing wood has only 80% of the $^{14}\mathrm{C}$ activity as found in living trees. The age of the artefact would be:
[Given: log 1.25 = 0.0969]

A) 1845 years **B)** 184.5 years
C) 1900 years **D)** 190 years

48. Fill in the blank :

$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow ? + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$

A) ${}^{141}_{56}\mathrm{Ba}$
B) ${}^{139}_{56}\mathrm{Ba}$
C) ${}^{139}_{54}\mathrm{Ba}$
D) ${}^{141}_{54}\mathrm{B}$

49. For the reaction,

$2N_2O_5 \rightarrow 4\ NO_2 + O_2$

the values of rate constant and rate of reaction are respectively $3 \times 10^{-5} s^{-1}$ and $2.4 \times 10^{-5} mol\ L^{-1} s^{-1}$. The concentration of $N_2O_5\,(\mathrm{in\ mol}\ L^{-1})$ will be:

A) 0.4 B) 0.8 C) 1.2 D) 0.6

50. Which is incorrect statement in the following?

A) α -rays have more penetrating power than β -rays
B) α -rays have less penetrating power than γ -rays
C) β -rays have less penetrating power than γ -rays
D) β -rays have more penetrating power than α -rays

51. The mechanism of the reaction, $\mathrm{2NO\,(g) + 2H_2\,(g) \rightarrow N_2\,(g) + 2H_2O\,(g)}$, is
Step 1: $\mathrm{2NO\,(g) + H_2\,(g) \xrightarrow{slow} N_2 + H_2O_2}$
Step 2: $\mathrm{H_2O_2 + H_2 \xrightarrow{fast} 2H_2O}$
Select the correct statement

A) rate $= \mathrm{k[NO]^2[H_2]^2}$
B) rate $= \mathrm{k\,[H_2O_2]\,[H_2]}$

D) if the initial concentration of $\mathrm{H_2}$ and NO is $\mathrm{C_0}$ and after time 't' the concentration of $\mathrm{N_2}$ is x, then rate $= \mathrm{k(C_0 - 2x)^x}$

52. The temperature coefficient of a reaction is 2. When the temperature is increases from $30°\mathrm{C}$ to $90°\mathrm{C}$, the rate of reaction is increased by

A) 150 times B) 410 times C) 72 times D) 64 times

53. When a bio-chemical reaction is carried out in laboratory, outside the human body in absence of enzyme, then rate of reaction obtained is 10^{-6} times, the activation energy of reaction in the presence of enzyme is

A) -6 RT B) +6 RT
C) -6 (2.303) RT D) +6 (2.303) RT

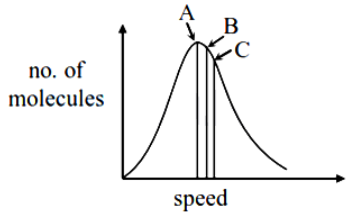
54. Which of the given radiations is having high penetrating power and not affected by electrical and magnetic field?

A) Alpha rays B) Beta rays
C) Gamma rays D) Neutrons

55. Calculate the average life (in minutes), if the half-life of a radionuclide is 69.3 minutes.

A) 100 B) $1/100$ C) 69.3×14.4 D) 0.693×69.3

56. Identify the correct labels of A, B and C in the following graph from the option given below



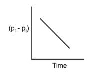
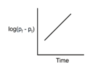
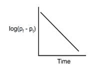
Root mean square speed $(\mathrm{V_{rms}})$; most probable speed $(\mathrm{V_{mp}})$; Average speed $(\mathrm{V_{av}})$

A) $\mathrm{A{-}V_{rms}; B{-}V_{mp}; C - V_{av}}$
B) $\mathrm{A - V_{mp}; B - V_{av}; C - V_{rms}}$
C) $\mathrm{A - V_{mp}; B - V_{rms}; C - V_{av}}$
D) $\mathrm{A - V_{av}; B - V_{rms}; C - V_{mp}}$

57. The half life period for catalytic decomposition of $\mathrm{XY_3}$ at 100 mm is found to be 8 hrs and at 200 mm it is 4 hrs. The order of reaction is

A) 3 B) 2 C) 4 D) None of these

58. Benzene diazonium chloride in aqueous solution decomposes as :
$\mathrm{C_6H_5 - N = N^+Cl^-_{(aq)} + H_2O_{(aq)} \longrightarrow C_6H_5OH_{(aq)} + N_{2(g)} + HCl_{(}}$
The reaction follows first order kinetics. If p_t is the pressure of N_2 at constant volume and temperature corresponding to different intervals of time t and p_f that after completion of the reaction, then which of the following graphs conforms to the kinetics of the reaction ?

A)  B)  C)  D)

59. Plotting $\mathrm{log_{10}t_{1/2}}$ against $\mathrm{log_{10}\,[A_0]}$ (where $\mathrm{A_0}$ is the initial concentration of a reactant) for a first order reaction the slope will be

A) -2 B) Zero C) $+1$ D) -1

60. The rate law for the chemical reaction

$\mathrm{2NO_2Cl \rightarrow 2NO_2 + Cl_2 = k\,[NO_2Cl]}$ is rate The rate determining step is

A) $\mathrm{2NO_2Cl \rightarrow 2NO_2 + 2Cl}$
B) $\mathrm{NO_2 + Cl_2 \rightarrow NO_2Cl + Cl}$
C) $\mathrm{NO_2Cl + Cl \rightarrow NO_2 + Cl_2}$
D) $\mathrm{NO_2Cl \rightarrow NO_2{+}Cl}$

61. For the cell reaction,
$\underset{(C_1)}{\mathrm{Cu^{2+}}}\mathrm{(aq) + Zn(s) \rightarrow} \underset{(C_2)}{\mathrm{Zn^{2+}}}\mathrm{(aq) + Cu(s)}$,
the change in free energy (ΔG) at a given temperature is a function of

A) $\mathrm{lnC_1}$ B) $\mathrm{ln\,(C_2/C_1)}$ C) $\mathrm{ln\,(C_1 + C_2)}$ D) $\mathrm{lnC_2}$

62. Three Faraday of electricity is passed through aqueous solutions of $AgNO_3, NiSO_4$ and $CoCl_3$ kept in three vessels using inert electrodes. The ratio in mol in which the metals Ag, Ni and Co will be deposited is

A) 3:2:1 B) 1:2:3 C) 2:3:6 D) 6:3:2

63. The equivalent conductivity of 0.1 M weak acid is 100 times less than that at infinite dilution. The degree of dissociation of weak electrolyte at 0.1 M is

A) 100 B) 10 C) 0.01 D) 0.001

64. An electrolytic cell is constructed for preparing hydrogen. For an average current of 2 ampere in the circuit, the time required to produce 450 mL of hydrogen at NTP is approximately

A) 1/2 hour B) 1 hour C) 2 hours D) 5 hours

65. The conductivity of strong electrolyte

A) Increases on dilution slightly
B) Decreases on dilution
C) Does not change with dilution
D) Depends upon density of electrolyte itself

66. Calculate the volume of O_2 liberated at STP by passing $5\mathrm{A}$ current for $193\ \mathrm{sec}$ through acidified water:

A) $56\ \mathrm{mL}$ B) $112\ \mathrm{mL}$
C) $158\ \mathrm{mL}$ D) $965\ \mathrm{mL}$

[illegible]

The E° for $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ will be :

A) $-2.69\ \mathrm{V}$; the reaction will not occur
B) $-2.69\ \mathrm{V}$; the reaction will occur
C) $-0.33\ \mathrm{V}$; the reaction will not occur
D) $-0.33\ \mathrm{V}$; the reaction will occur

68. Mark out the correct statement among the following.

A) Aqueous $AgNO_3$ solution can be stored in a copper bowl.
B) Aqueous $CuSO_4$ solution can be stored in a silver bowl.
C) Cu , Ag can release hydrogen gas from dil HCl .
D) H_2 cannot reduce Cu^{2+} and Ag^{+} in the form of metals.

69. The equivalent conductance of $1\mathrm{M}$ benzoic acid is $12.8\mathrm{ohm}^{-1}\mathrm{cm}^2$ and if the conductance of benzoate ion and H^{+} ion at infinite dilution are 42 and $288.42\mathrm{ohm}^{-1}\mathrm{cm}^2$ respectively. Then its degree of dissociation is

A) 0.039% B) 3.9% C) 0.35% D) 39%

70. A silver cup is plated with silver by passing 965 C of electricity. The amount of Ag deposited is

A) 107.89 g B) 9.89 g C) 1.0002 g D) 1.08 g

71. The EMF of a cell corresponding to the reaction :

$$\mathrm{Zn(s)} + 2\mathrm{H}^{+}(\mathrm{aq}) \rightarrow \mathrm{Zn}^{2+}(0.1\ \mathrm{M}) + \mathrm{H_2(g)}\ (1\ \mathrm{atm.})\ \text{is}\ 0.28\ \text{volt}$$

The pH of the solution at the hydrogen electrode.

$$\mathrm{E}^{\mathrm{o}}_{\mathrm{Zn^{2+}}\,/\,\mathrm{Zn}} = -0.76\ \text{volt};\ \mathrm{E}^{\mathrm{o}}_{\mathrm{H^{+}}\,/\,\mathrm{H_2}} = 0$$

A) 2.30 B) 7.8 C) 9.2 D) 8.30

72. 200 ml of water is added to 500 ml of 0.2 M solution. What is the molarity of this diluted solution?

A) 0.7093 M B) 0.1428 M C) 0.4005 M D) 0.2897 M

73. The reactions that occurs at the cathode of a common dry cell is

A) $2ZnO_2 + Mn^{2+} + 2e^{-} \rightarrow MnZn_2O_4$
B) $Mn \rightarrow Mn^{2+} + 2e^{-}$
C) $2MnO_2 + Zn^{2+} + 2e^{-} \rightarrow ZnMn_2O_4$
D) $Zn \rightarrow Zn^{2+} + 2e^{-}$

74. Given that E° values of $Ag^{+}/Ag,\ K^{+}/\ K,\ Mg^{2+}/Mg$ and Cr^{3+}/Cr are $0.80\mathrm{V},\ -2.93\mathrm{V}, -2.37\mathrm{V}$ and $-0.74\mathrm{V}$ respectively. Therefore the order for the reducing power of the metal is -

A) $\mathrm{Ag} > \mathrm{Cr} > \mathrm{Mg} > \mathrm{K}$
B) $\mathrm{Ag} < \mathrm{Cr} < \mathrm{Mg} < \mathrm{K}$
C) $\mathrm{Ag} > \mathrm{Cr} > \mathrm{K} > \mathrm{Mg}$
D) $\mathrm{Cr} > \mathrm{Ag} > \mathrm{Mg} > \mathrm{K}$

75. The material used to make anode and cathode of the commonly employed dry cells are

A) aluminium and lead
B) zinc and carbon
C) an alloy of silver, copper and manganese
D) brass and a mixture of ammonium chloride and manganese

76. The vapour pressure of pure benzene at $25\,°\mathrm{C}$ is 640 mm Hg and that of the solute A in benzene is 630 mm of Hg. The molality of solution is

A) 0.2 m B) 0.4 m C) 0.5 m D) 0.1 m

77. 0.01 M solution of KCI and $CaCl_2$ are separately prepared in water. The freezing point of KCI is found to be $-2°\mathrm{C}$. What is the freezing point of $CaCl_2$ aq. solution if it is completely ionized?

A) $-3°\mathrm{C}$ B) $+\ 3°\mathrm{C}$ C) $-2°\mathrm{C}$ D) $-4°\mathrm{C}$

78. Which substance would give a solution with a boiling point below that of pure water rather than above ?

A) Sodium chloride (solid)
B) Ethyl alcohol (liquid, b.p. 61°C)

C) Sulphuric acid (liquid, b.p. > 300°C)
D) Sucrose sugar (solid)

79. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of the $Al_2(SO_4)_3$ (if all are 100% ionised)?

A) K_2SO_4 B) $K_3[Fe(CN)_6]$
C) $K_4[Fe(CN)_6]$ D) $Al(NO_3)_3$

80. At 35°C, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is

A) A mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL
B) Roult's law is not obeyed by this system
C) Heat must be absorbed in order to produce the solution at 35°C
D) CS_2 and acetone are less attracted to each other than to themselves

81. The vapour pressure of pure liquid A decreases from 0.80 atm to 0.6 atm on mixing a non-volatile solute B to liquide A . The mole fraction of B in the solution is

A) 0.25 B) 1 C) 0.50 D) 0.75

82. Urea $(NH_2 - CO - NH_2)$ needs to be dissolved in 100g of water, in order to decrease the vapour pressure of water by 25%. What will be the molality of the solution?

in freezing point is lowest?

A) 0.2 M urea and 0.2 M glucose
B) 0.1 M $Al_2(SO_4)_3$ and 0.1 M Na_2SO_4
C) 0.1 $MKNO_3$ and 0.2 M $Ba(NO_3)_2$
D) 0.1 M $Ca(NO_3)_2$ and 0.1 M $Ba(NO_3)_2$

84. Dissolution of 1.5 g of non-volatile solute (Molecular weight = 60) in 250 g of a solvent reduces its freezing point by $0.01^\circ C$. Find the molal depression constant of the solvent.

A) 0.01 B) 0.001 C) 0.0001 D) 0.1

85. Insulin $(C_6H_{10}O_5)_n$ dissolved in a medium shows osmotic pressure π at C gm/ cm^3 concentration and 300 K temperature. The slope of plot of π against C is 4.1×10^{-3} . Molecular mass of insulin is

A) 3×10^3 B) 6×10^6 C) 3×10^6 D) 6×10^3

86. Which one of the following statements regarding Henry's law is not correct?

A) The value of K_H changes with function of the nature of the gas.
B) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids
C) The partial pressure of the gas in vapour phase is proportional to the mole fraction of of the gas in the solution.
D) Different gases have different K_H (Henry's law constant) value at a same temperature.

87. For a dilute solution containing 2.5 g of a non-volatile non–electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is $2^\circ C$. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (in mm of Hg) of the solution is (take $K_b = 0.76$ Kkg mol^{-1})

A) 718 B) 736 C) 724 D) 740

88. The vapour pressure of water at $20^\circ C$ is 17.5 mmHg.

If 18 g of glucose $(C_6H_{12}O_6)$ is added to 178.2 g of water at $20^\circ C$, the vapour pressure of the resulting solution will be

A) 17.675 mmHg B) 15.750 mmHg
C) 16.500 mmHg D) 17.325 mmHg

89. The factors on which degree of ionization of a compound depends is

A) Size of solute molecules
B) Quantity of electricity passed
C) Nature of vessel used
D) Nature of solute molecules

90. pH of 0.1 M BOH (weak base) is found to be 12. The solution at temperature T K will display an osmotic pressure equal to:

A) 0.01 RT B) $0.01(RT)^2$ C) 0.11 RT D) 1.1 RT

Biology - (Zoology)

91. Which of the following can be complications resulting from $STDs$ without timely detection and treatment?
A. Pelvic inflammatory disease (PID)
B. Still births
C. Infertility or even cancer of reproductive tract
D. Ectopic pregnancies

A) A & B only
B) A, B & C
C) A only
D) A, B, C & D

92. Match List I with List II

List I	List II
A. Non-medicated IUD	I. Multiload 375
B. Copper releasing IUD	II. Progestogens
C. Hormone releasing IUD	III. Lippes loop
D. Implants	IV. $LNG-20$

Choose the correct answer from the option given below:

A) $A-I, B-III, C-IV, D-II$
B) $A-IV, B-I, C-II, D-III$
C) $A-III, B-I, C-IV, D-II$
D) $A-III, B-I, C-II, D-IV$

93. Which of the following $STDs$ cannot be treated with antibiotics?
$A.$ Gonorrhoea $B.$ Syphilis $C.$ Chlamydia $D.$ Genital herpes

A) $D\ only$ **B)** $B\ and\ D$
C) $C\ and\ D$ **D)** $C\ only$

94. Match the following columns

Column$-I$	Column$-II$
(A) Chlamydiasis	(1) Chlamydia trachomatis
(B) Gonorrhoea	(2) Neisseria gonorrhoeae
(C) Trichomoniasis	(3) Trichomonas vaginalis
(D) Genital herpes	(4) Herpes simplex virus
(E) Syphilis	(5) Treponema pallidum

A) $(A)-(2).\ (B)-(5),\ (C)-(3),\ (D)-(1),\ (E),\ (4)$
B) $(A)-(4).\ (B)-(5),\ (C)-(3),\ (D)-(2),\ (E),\ (5)$
C) $(A)-(1).\ (B)-(2),\ (C)-(3),\ (D)-(4),\ (E),\ (5)$
D) $(A)-(1).\ (B)-(2),\ (C)-(4),\ (D)-(3),\ (E),\ (5)$

95. Misuse of amniocentesis results in

A) Blood Mismatch **B)** Death Of Foetus
C) Genetic Disorders **D)** None of these

96. Our laws permit legal .... and it is as yet, one of the best methods for couples looking for

the highest failure rate?

A) Diaphragm with spermicide **B)** Condom
C) Intrauterine device **D)** Rhythm method

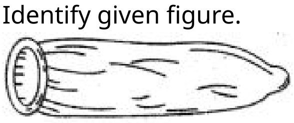
98. Implants under the skin and injectables contain

A) Progestogen alone
B) Progestogen and estrogen
C) $FSH\ and\ LH$
D) Both (a) & (b) are correct

99. 'family planning' program was initiated in ....

A) 1950 **B)** 1951 **C)** 1952 **D)** 1953

100. Identify given figure.



A) Condom for female **B)** Loop
C) CuT **D)** Condom for male

101. What is the full form of $ZIFT$?

A) Zygote Induce Fallopian Transfer
B) Zygote Intra Fallopian Transfer
C) Zygote Inert Fallopian Transfer
D) Zygote In Fallopian Transfer

102. Which oral contraceptive is developed by $CDRI$?

A) Saheli **B)** Mala$-D$
C) Both (a) and (b) **D)** None of these

103. What is the purpose of surgical method of contraception?

A) Prevent gamete motility
B) Prevent gamete formation
C) Gametogenesis promotion
D) Facilitate implantation

104. Ovulation do not occur in lactational period because of

A) Inhibin **B)** Prolactin
C) Prostaglandin **D)** Oxytocin

105. Identify the correct statements
$i.$ Birth control pills are likely to cause cardiovascular problem
$ii.$ A woman who substitutes or takes the place of the real mother to nurse to embryo is called surrogate mother
$iii.$ Numerous children have been produced by in vitro fertilisation but with some abnormalities
$iv.$ Woman plays a key role in the continuity of the family and human species
$v.$ Foetal sex determination test should not be banned

A) I and II
B) II and IV
C) III and V
D) I, II and IV

106. Which of the following is an incorrect statement for periodic abstinence?

A) The couple should abstain from coitus from day 10 to 17 of the menstrual cycle when ovulation could be expected
B) 10^{th} to 17^{th} day of the cycle is fertile period, when the chances of fertilisation are high
C) This prevents the chances of union of male and female gametes
D) In this method, the ovum and sperms are prevented from physically meeting with the help of barriers

107. Which of the following is world's first non-hormonal oral contraceptive pill for females, developed by scientists at Central Drug Research Institute $(CDRI)$ in Lucknow, India?

A) Mala$-D$ **B)** Saheli
C) Morning after pills **D)** PoP

108. Birth rate is directly proportional to

A) Good food **B)** Illiteracy
C) High social status **D)** Age of marriage

109. One of the legal methods of birth control is

A) by having coitus at the time of day break
B) by a premature ejaculation during coitus
C) by abstaining from coitus from day 10 to 17 of the menstrual cycle.

D) abortion by taking an appropriate medicine

110. At present total population of the world is about

A) 4.5 billions B) 5.2 billions
C) 6.2 billions D) None of the above

111. Trade name of weekly oral contraceptive pill is

A) Mala B) Saheli C) Mala A D) Mala D

112. Tubectomy is a method of sterilization in which

A) small part of the Fallopian tube is removed or tied up
B) ovaries are removed surgically
C) small part of vas deferens is removed or tied up
D) uterus is removed surgically.

113. Which gland's secretion is act as lubricants in male?

A) Bulbourethral gland B) Pancrease
C) Seminal vesicle D) Vas deferens

114. Which organ of human is not paired ?

A) Vas deferens B) Prostate gland
C) Bulbourethral gland D) Testis .

115. Which of the following hormones is not a secretory product of human placenta?

A) Human chorionic gonadotropin B) Prolactin

the menstrual cycle because

A) Follicles do not remain in the ovary after ovulation
B) FSH levels are high in the luteal phase
C) LH levels are high in the luteal phase
D) Both FSH and LH levels are low in the luteal phase

117. Ejaculation of human male contains about $200 - 300$ million sperms, of which for normal fertility ____ $\%$ sperms must have normal shape and size and at least ____$\%$ must show energetic motility.

A) $40, 60$ B) $50, 50$
C) $60, 40$ D) $30, 70$

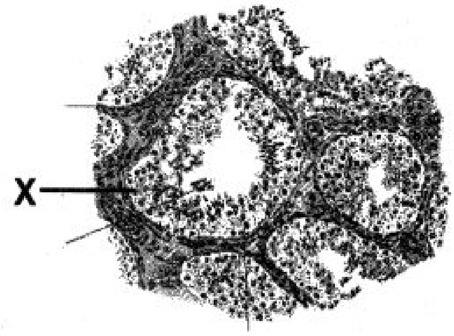
118. The correct sequence of spermatogenetic stages leading to the formation of sperms in a mature human testis is

A) spermatogonia - spermatocyte - spermatid - sperms
B) spermatid - spermatocyte - spermatogonia - sperms
C) spermatogonia - spermatid - spermatocyte - sperms
D) spermatocyte - spermatogonia - spermatid - sperms.

119. In human adult females oxytocin

A) stimulates pituitary to secrete vasopressin
B) causes strong uterine contractions during parturition
C) is secreted by anterior pituitary
D) stimulates growth of mammary glands.

120. Identify X from figure



A) Interstitial cells B) Spermatogonia
C) Langerhans cell mass D) Sertoli cells

121. If the vas deferens of a man is surgically disconnected

A) Sperms in the semen will be without nuclei
B) Semen will be without sperms
C) Spermatogenesis will not occur
D) Sperms in the semen will be non-motile

122. The placenta helps in the maintenance of pregnancy by secreting a hormone, known as

A) Thyroxine B) Estrogen
C) Progesteron D) Testosterone

123. If after ovulation no pregnancy results, the corpus luteum

A) Is maintained by the presence of progesterone
B) Degenerates in a short time
C) Becomes active and secretes lot of FSH and LH
D) Produces lot of oxytocin and relaxin

124. The process by which ova are formed is knwon as

A) Oogenesis B) Ovulation
C) Oviposition D) Oviparity

125. Fertilization is depicted by the condition

A) $n \to 2n$ B) $2n \to 3n$
C) $2n \to 4n$ D) $4n \to 8n$

126. Which one of the following statements with regard to embryonic development in humans is correct

A) Cleavage division bring about considerable increase in the mass of protoplasm
B) In the second cleavage division, one of the two blastomeres usually divides a little sooner than the second
C) With more cleavage divisions, the resultant blastomeres become larger and larger

D) Cleavage division results in a hollow ball of cells called morula

127. Gastrulation is the process which involves the differentiation of the following layers in a vertebrate embryo

A) Ectoderm and mesoderm
B) Ectoderm and endoderm
C) Endoderm and mesoderm
D) Ectoderm, endoderm and mesoderm

128. Ovulation occurs

A) Alternately from two ovaries
B) Simultaneously from both the ovaries
C) From one ovary alone throughout the life
D) According to the season from two ovaries

129. The male sex hormone 'testosterone' is secreted by special Leydig's cells present in

A) Kidneys
B) Seminiferous tubules of testes
C) Connective tissue surrounding the seminiferous tubules of testes
D) Cells lining the epidermis of the testes

130. Functions of seminal fluid is/are

A) Maintains the viability of sperms
B) Maintains motility of sperms
C) Provides proper pH and ionic strength
D) All the above

131. The endometrium is the lining of

A) Bladder B) Vagina C) Uterus D) Oviduct

C) Vagina and clitoris D) Clitoris and labia

133. The cavity present in the graafian follicle is

A) Amniotic cavity B) Archenteron
C) Antrum D) Ostium

134. Fertilization occurs in human, rabbit and other placental mammals in

A) Ovary B) Uterus C) Fallopian tubes D) Vagina

135. During the course of development, cells in various regions of embryo become variable in morphology and eventually perform diverse functions. This process is known as

A) Rearrangement B) Differentiation
C) Metamorphosis D) Organisation

Biology - (Botany)

136. Scutellum is

A) Cotyledon in dicots
B) Cotyledon in gymnosperm
C) Monocot root
D) Cotyledon in grass family

137. Parthenogenesis is a type of

A) Sexual reproduction
B) Asexual reproduction
C) Budding
D) Regeneration

138. Function of micropyle is

A) Helps in germination
B) Helps in surviving
C) Both (a) and (b)
D) Helps in endosperm formation

139. The innermost wall layer of anther

A) Is nutritive in function
B) Helps in dehiscence of anther
C) Is haploid and protective in function
D) Forms microspores

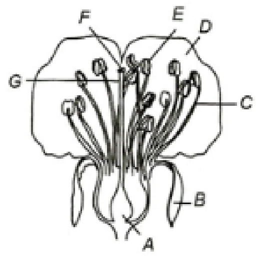
140. In $40\ \%$ angiosperms, the pollen grains are shed at

A) Four-celled stage B) Three-celled stage
C) Two-celled stage D) Five-celled stage

141. The pollen viability period of rice and pea respectively, is

A) $30\ minutes$ and several months
B) Several months and $30\ minutes$
C) Few days and few months
D) Few days in both the cases

142. Identify A to G in following figure and answer accordingly



A) A—Ovary, B—Filament, C—Sepal, D—Petal, E—Style, F—Stigma, G—Anther
B) A—Petal, B—Ovary, C—Petal, D—Filament, E—Anther, F—Stigma, G—Style
C) A—Ovary, B— Sepal, C— Filament, D— Petal, E—Anther, F—Stigma, G—Style
D) A—Ovary, B—Sepal, C— Filament, D— Petal, E—Anther, F—Stigma, G—Style

143. Sexual reproduction leads to

A) Genetic recombination B) Polyploidy
C) Aneuploidy D) euploidy

144. Milky water of tender coconut is

A) Liquid gametes
B) Liquid nucellus
C) Liquid female gametophyte
D) Liquid endosperm

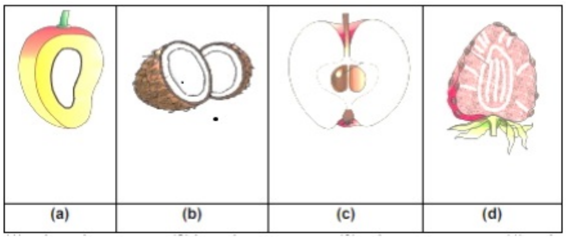
145. The number of female nuclei involved in double fertilization is

A) 2 B) 3 C) 4 D) 1

146. Flower is a

A) Modified male plant only
B) Modified female plant only
C) Modified reproductive shoot
D) Vegetative shoot system

147. Identify which of the following fruits are false fruit?



A) a, b, c, d B) b, c, d
C) a, b, c D) c, d

148. Which is incorrect statement?

I. Each cell of sporogenous tissue in anther is capable of giving rise to microspore tetrad.
II. The pollen grain represent male gametophyte.
III. Pollen grains are usually triangular and $10-15 \mu m$ in diameter.
IV. Sporopollenin is one of the most resistance organic material which can be destroyed only by strong acids
and alkali.

A) I, II are incorrect but III, IV are correct
B) III, IV are incorrect but I, II are correct

A) Outer three layers of anther wall are protective in function
B) Sporogenous tissue, occupies the centre of each microsporangium
C) Cells of tapetum and endothecium show increase in DNA contents by endomitosis and polyteny
D) Ploidy level of microspore tetrad is haploid

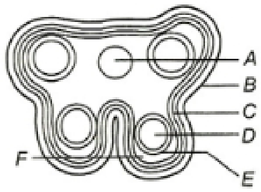
150. Which of the following plant provides safe place to insect for laying eggs?

A) Sage plant B) Amorphophallus
C) Ophrys D) Mango

151. What does the filiform apparatus do at the entrance into ovule?

A) It helps in the entry of pollen tube into a synergid
B) It prevents entry of more than one pollen tube into the embryo sac
C) It brings about opening of the pollen tube
D) It guides pollen tube from a synergid to egg

152. Identify A to E in the following diagram

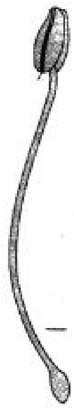


A) A–Epidermis, B–Endodermis, C–Connective tissues, D–Sporogenous tissue, E–Middle layer, F–Tapetum
B) A–Endodermis, B– Connective tissues, C–Epidermis, D– Tapetum, E–Sporogenous tissue, F– Middle layer
C) A– Tapetum, B– Middle layer, C–Sporogenous tissue, D– Connective tissues, E–Endodermis, F–Epidermis
D) A– Connective tissues, B– Epidermis, C–Endothecium, D–Sporogenous tissue, E–Tapetum, F– Middle layer

153. Which one of the following statements is correct?

A) Endothecium produces the microspores.
B) Tapetum nourishes the developing pollen.
C) Hard outer layer of pollen is called intine.
D) Sporogenous tissue is haploid.

154. Identify figure.



A) Typical gynoecium B) Typical stamen
C) Cut section of microsporangium D) Flower

155. Which one of the following is resistant to enzyme action?

A) Pollen exine B) Leaf cuticle
C) Cork D) Wood fibre

156. Nucellar polyembryony is reported in species of

A) Citrus B) Gossypium C) Triticum D) Brassica.

157. Both, autogamy and geitonogamy are prevented in

A) papaya B) cucumber C) castor D) maize.

158. Which one of the following statements is correct?

A) Cleistogamous flowers are always autogamous.
B) Xenogamy occursonly bywind pollination.
C) Chasmogamous flowers do not open at all.

D) Geitonogamy involves the pollen and stigma of flowers of different plants.

159. Where does pollen tube developes ?

A) In style
B) In anthor
C) On the stigma
D) In stalk (Filment of anthor)

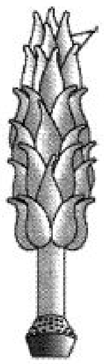
160. In which of the plants the seeds are thousands?

A) Phoenix dectylyfera B) Mango
C) Orchids D) Carrot

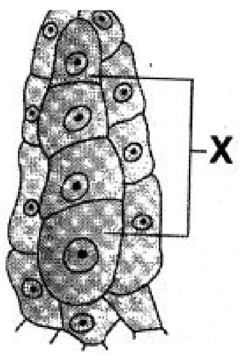
161. It is not included in post fertilisation event

A) Seed formation B) Fruit formation
C) Gametogenesis D) Embryogenesis

162. Identify the given figure :



A) Flower of michelia
B) Gynoecium of michelia



A) PMC B) Megaspore Tetrad
C) Microspore D) Megaspore

164. ....pollination is quite common in grasses

A) Water B) Wind C) Insect D) Animal

165. Pollination in water hyacinth and water lily is brought about by the agency of

A) water B) insects or wind C) birds D) bats.

166. The endosperm of Brassicais

A) Haploid B) Diploid C) Triploid D) Tetraploid

167. Fertilization of egg takes place inside

A) Anther B) Stigma C) Pollen tube D) Embryo sac

168. Cheiropterophily is the process of pollination by

A) Water B) Bat C) Insect D) Bird

169. Which of the following is without exception in angiosperms

A) Secondary growth
B) Presence of vessels
C) Double fertilization
D) Autotrophic nutrition

170. In a seed of maize, scutellum is considered as cotyledon because it

A) Protects the embryo
B) Contains food for the embryo
C) Absorbs food materials and supplies them to the embryo
D) Converts itself into a monocot leaf

171. The sequence of development of embryo sac is

A) Archesporium $\rightarrow$ megaspore mother cell $\rightarrow$ megaspore $\rightarrow$ embryo sac
B) Archesporium $\rightarrow$ megaspore $\rightarrow$ megaspore mother cell $\rightarrow$ embryo sac
C) Archesporium $\rightarrow$ megaspore $\rightarrow$ megasporophyte $\rightarrow$ embryo sac
D) None of the above

172. The pollen tube usually enters the embryo sac

A) Between one synergid and central cell
B) By directly penetrating the egg
C) Through one of the synergids
D) By knocking off the antipodal cells

173. Which of the following pairs of plant parts are both haploid

A) Antipodal cells and egg cells
B) Nucellus and primary endosperm nucleus
C) Nucellus and antipodal cells
D) Antipodal cells and megaspore mother cells

174. The stalk of the ovule is called

A) Pedicel B) Petiole C) Funicle D) Hilum

175. In which type whole of the megaspore mother cell takes part in the formation of the female gametophyte

A) Monosporic 8 nucleate
B) Monosporic 4 nucleate
C) Bisporic
D) Tetrasporic

176. The mature stigma is either rough or sticky in

A) All types of flowers
B) Water pollinated flowers
C) Wind pollinated flowers
D) Insect pollinated flowers

177. Correct definition of pollination is

A) Transfer of pollen grain from anther to stigma
B) Germination of pollen grain

C) Growth of pollen tube in ovule
D) Visits of insects in flower

178. In which characters air pollinated flowers differ from insect pollinated ones

A) Due to small parianth and sticky pollen
B) Small coloured parianth and heavy pollen grains
C) Coloured parianth and large pollen grains
D) Without parianth and light pollen grains

179. Anther is generally composed of

A) One sporangium B) Two sporangium
C) Three sporangium D) Four sporangium

180. During meiosis in pollen mother cell the daughter cells are interconnected by passages. The whole structure is called

A) Symplast B) Plasmodesmata
C) Syncytium D) Coenocyte

Mitoprep

Sapno Ki Udaan